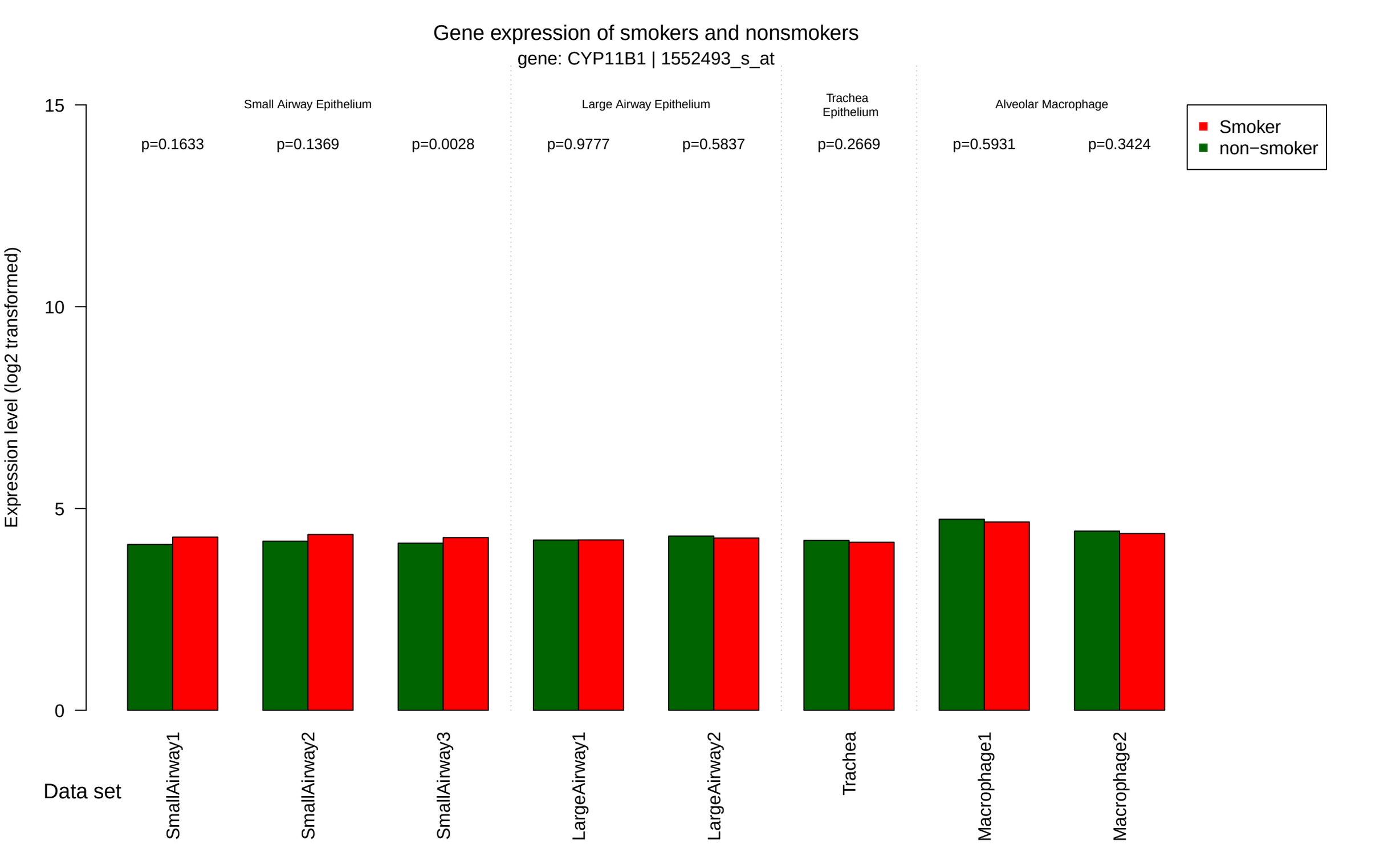
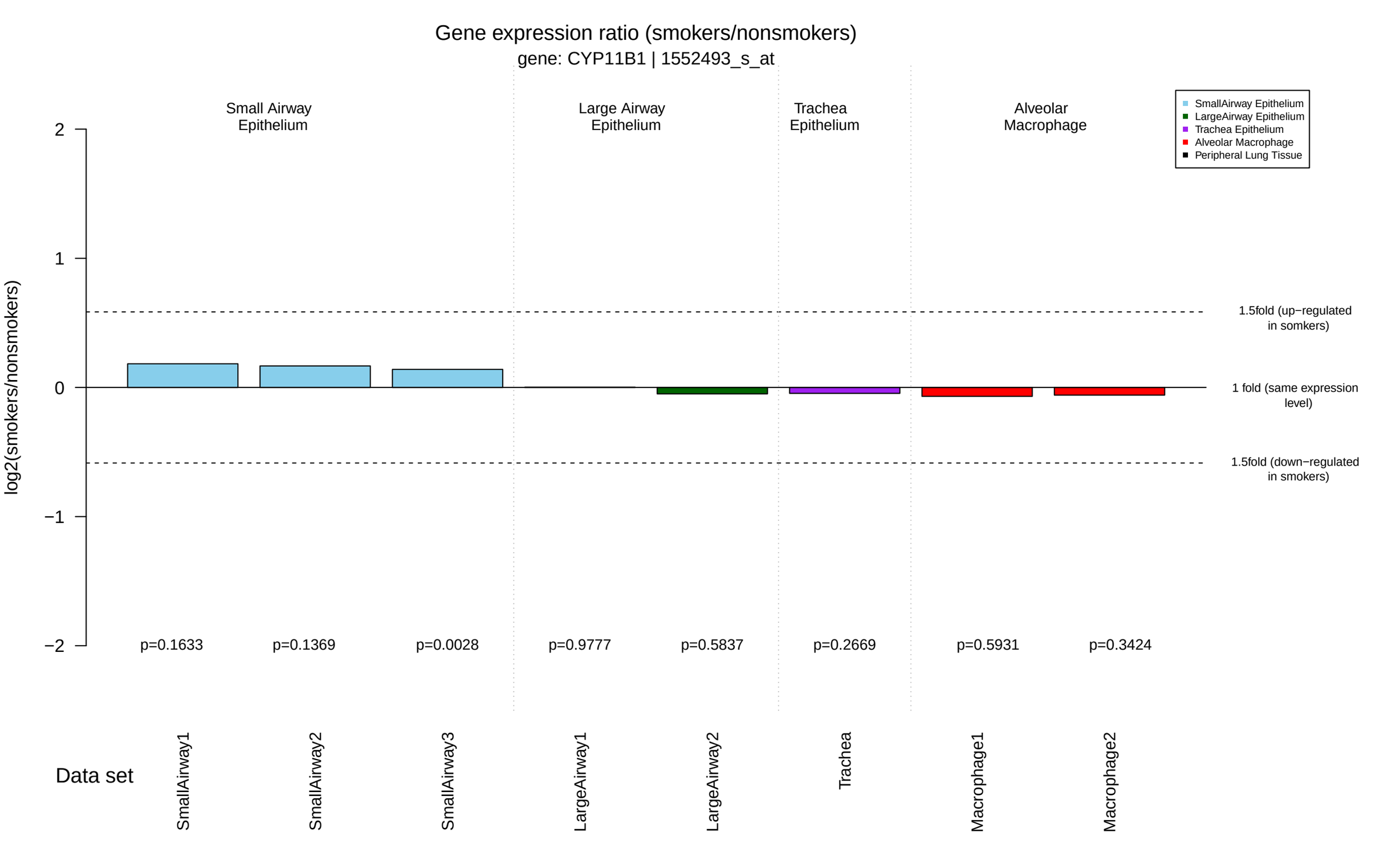
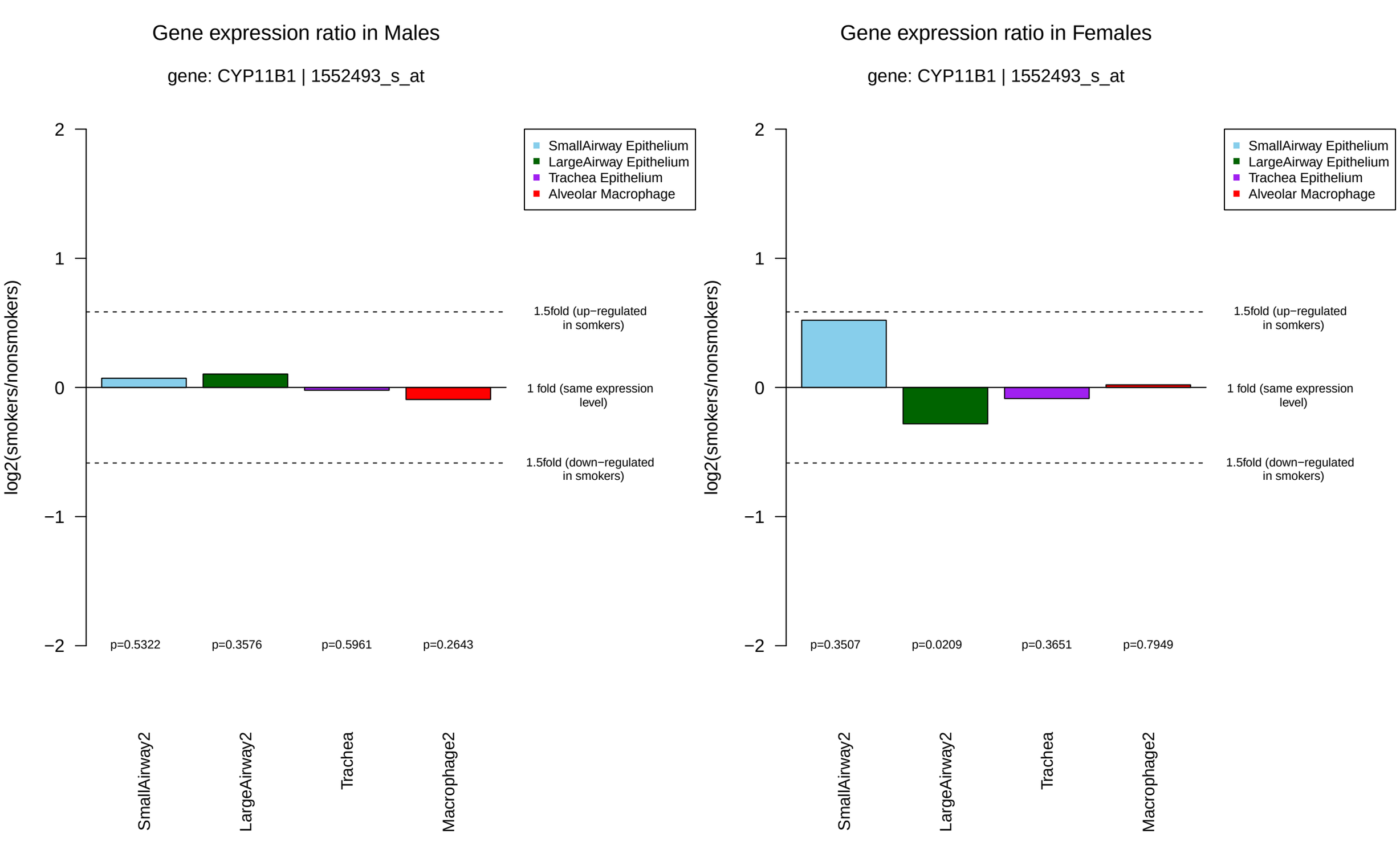
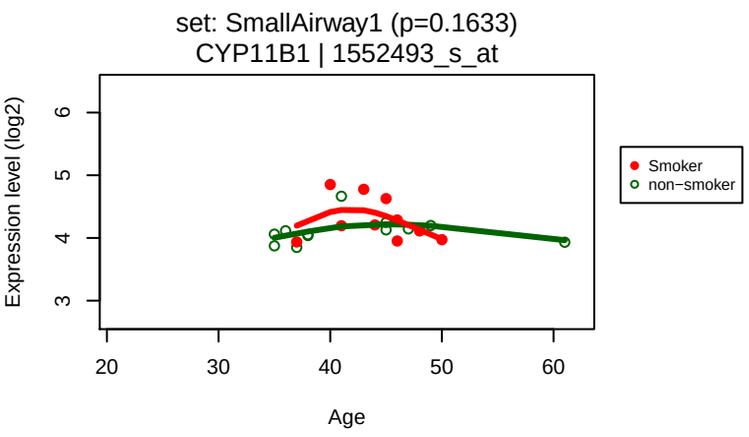
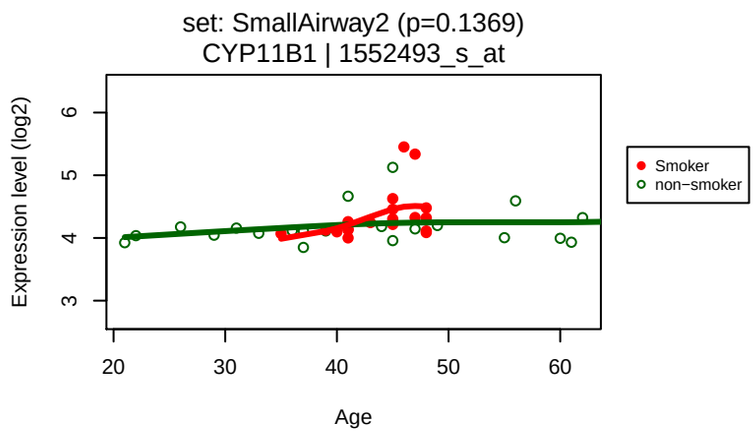
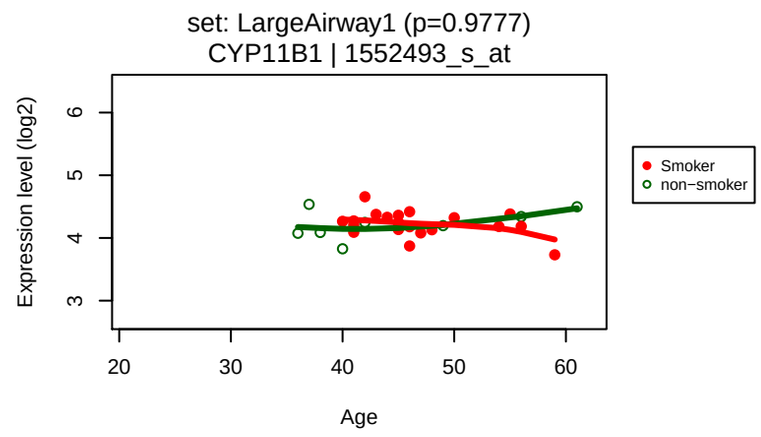

Gene expression of smokers and nonsmokers
gene: CYP11B1 | 1552493_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1633
p=0.1369
p=0.0028
p=0.9777
p=0.5837
p=0.2669
p=0.5931
p=0.3424
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

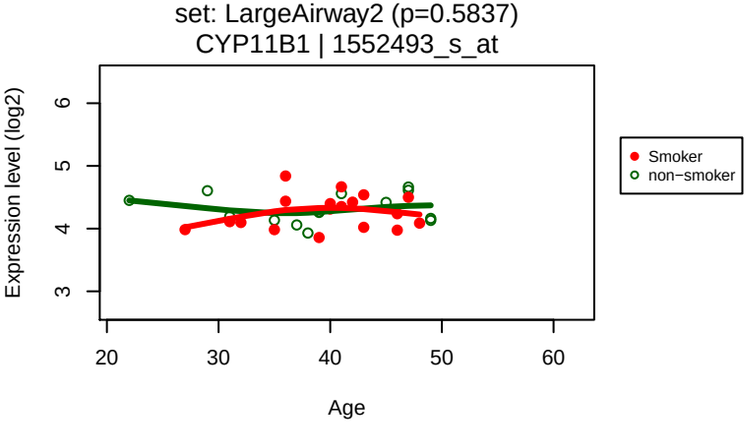

Gene expression ratio (smokers/nonsmokers)
gene: CYP11B1 | 1552493_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1633
p=0.1369
p=0.0028
p=0.9777
p=0.5837
p=0.2669
p=0.5931
p=0.3424
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

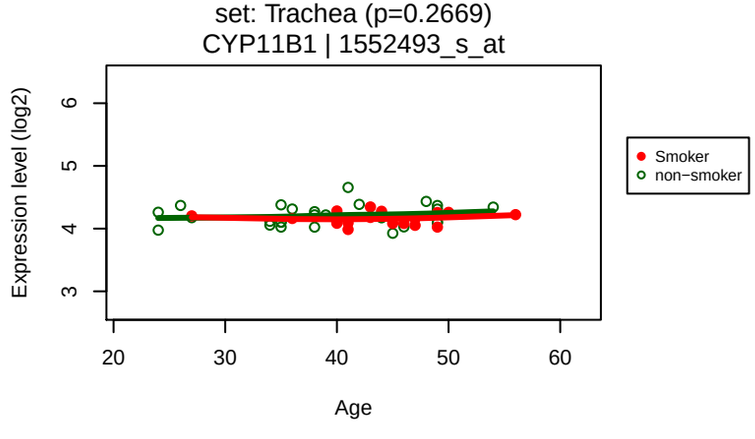

Gene expression ratio in Males
gene: CYP11B1 | 1552493_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5322
p=0.3576
p=0.5961
p=0.2643
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP11B1 | 1552493_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3507
p=0.0209
p=0.3651
p=0.7949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

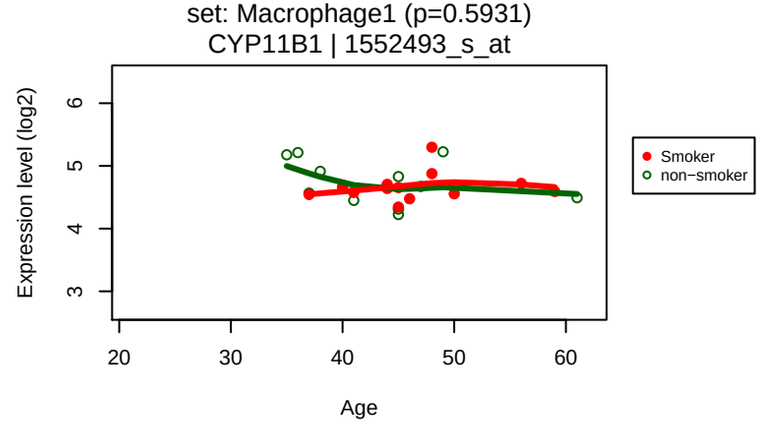

set: SmallAirway1 (p=0.1633)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker

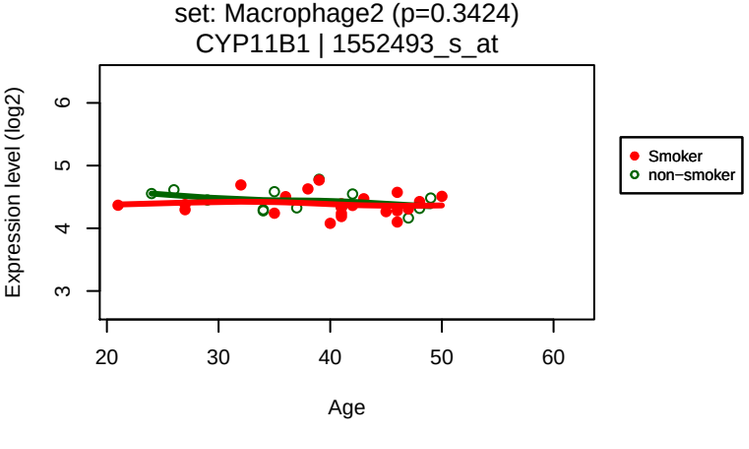

set: SmallAirway2 (p=0.1369)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker

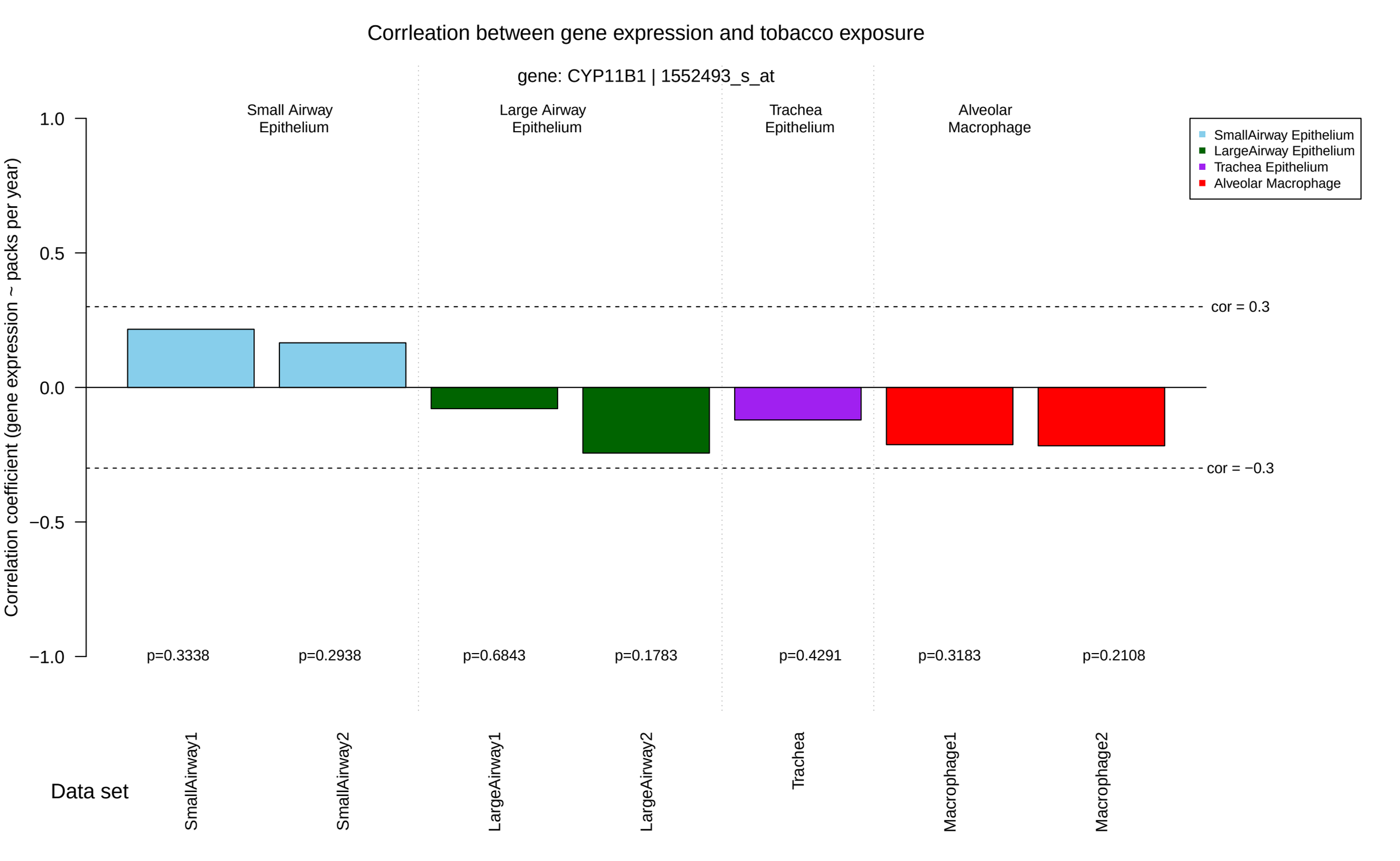

set: LargeAirway1 (p=0.9777)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5837)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2669)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5931)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3424)
CYP11B1 | 1552493_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP11B1 | 1552493_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3338
p=0.2938
p=0.6843
p=0.1783
p=0.4291
p=0.3183
p=0.2108
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2